

Harsh Tyagi

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Summary

Software Development Engineer with approximately one year of hands-on experience in Full-Stack development and Applied AI, contributing to production-grade SaaS platforms and AI-driven systems. Experienced in integrating Computer Vision models and LLM-based pipelines into scalable backend services using React.js, Node.js, Laravel, Python, and FastAPI. Strong exposure to the full development lifecycle, including data preparation, model training, API integration, and user-facing feature development.

Education

R.D. Engineering College (RDEC) <i>Bachelor of Technology in Computer Science (Artificial Intelligence & Machine Learning)</i> CGPA: 8.05 / 10.00	August 2021 – June 2025 Ghaziabad, Uttar Pradesh
Mount Carmel School, Muradnagar <i>Senior Secondary (Class XII) – Physics, Chemistry, Mathematics</i> CGPA: 7.50 / 10.00	August 2021 Ghaziabad, Uttar Pradesh

Experience

Redian Software Global Pvt. Ltd. <i>Software Development Engineer</i> - Designed, developed, and maintained scalable web applications using PHP and Laravel, implementing routing, middleware, authentication, and optimized database schemas to support production-grade workloads. - Contributed to AI-driven construction technology platforms, including an automated quantity take-off system and an AI-based tender prediction solution, covering end-to-end SaaS development from model training to deployment. - Trained and integrated custom AI/ML models using Python, including image-based models for domain-specific datasets (room windows, floor plans), and integrated inference results into web applications via APIs. - Applied LLM prompt engineering and AI workflow automation to enhance prediction accuracy and user interaction within AI-powered modules. - Worked across a full-stack environment using PHP (Laravel), React.js, Node.js, Express.js, MongoDB, SQL, Python (AI/ML), Computer Vision, Bootstrap, and Tailwind CSS.	March 2025 – Present Noida Sector 63, Uttar Pradesh
MaiVin Consulting Services Pvt. Ltd. <i>Software Engineer Intern</i> - Worked extensively with SQL and SAP HANA SQL to query, manage, and analyze enterprise data using in-memory database capabilities. - Developed business and analytical reports using Crystal Reports, supporting decision-making for SAP Business One implementations. - Gained hands-on experience with SAP Business One, database-driven reporting, and backend data workflows, alongside foundational Java development exposure.	July 2024 – December 2024 Greater Noida, Uttar Pradesh

Projects

AI Takeoff & Estimation Platform for Construction <i>React.js, Node.js, MongoDB, Python, FastAPI, LLMs, Computer Vision</i> - Developed an AI-powered construction takeoff and estimation platform enabling automated OCR extraction, scale detection, and accurate measurement from PDF and CAD drawings using custom-trained YOLO-based detection and segmentation models. - Implemented domain-specific computer vision pipelines for detecting rooms, windows, and architectural elements, including manual scale override logic for drawings with missing or inconsistent scale information. - Built a full SaaS architecture with user authentication, role-based access control, subscription workflows, and real-time estimation insights using React.js, Node.js, MongoDB, and Python-based FastAPI inference services.
AI Tender Management & Document Intelligence System <i>React.js, Node.js, MongoDB, Python, FastAPI, LLMs, Prompt Engineering</i> - Designed and implemented a SaaS-based AI tender and document management system supporting multi-company document storage, tender uploads across multiple file formats, and secure data access. - Built LLM-driven analysis pipelines to extract key tender parameters (budget, timelines, eligibility criteria, requirements) and predict tender winning probability using prompt-engineered workflows.

- Enabled end-to-end tender processing with real-time analysis tracking and automated report generation in PDF, Excel, and Word formats via scalable React.js and Node.js backend services.

ABC HRMS | *PHP, Laravel, SQL, Bootstrap, Tailwind CSS, Prompt Engineering*

- Developed a web-based Human Resource Management System to streamline employee lifecycle operations including attendance, leave management, internal communication, and HR workflows.
- Implemented secure authentication, role-based access, and database-driven modules using PHP (Laravel) and SQL, with a responsive UI built using Bootstrap and Tailwind CSS.

Technical Skills

Programming Languages: C, Java, Python, JavaScript (ES6+), PHP, SQL

Web & Backend Technologies: HTML5, CSS3, Bootstrap, Tailwind CSS, React.js, Node.js, Express.js, PHP (Laravel), REST APIs

Databases: MongoDB, MySQL

AI / Machine Learning: Applied Artificial Intelligence, Machine Learning, Neural Networks, Computer Vision, Image Annotation, Dataset Curation, Custom Model Training, Domain-Specific Models (Room Windows, Floor Plans, etc), Model Evaluation, Model Integration

AI Tooling & Automation: LLM Prompt Engineering, n8n Workflow Automation, AI Pipeline Integration

Security & Backend Concepts: JWT-Based Authentication, Encryption & Decryption, API Design, Database Normalization

Data & BI Tools: SAP Business One, Crystal Reports

Version Control & Tools: Git, GitHub

Certifications

- AWS Cloud Practitioner – Foundations
- Google Cloud Computing Foundations
- Meta (Coursera) – Version Control
- HackerRank – Java (4)
- Applied Machine Learning & Computer Vision (Project-Based Experience)